

Assignment 3- CP363

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Group # 26

Equipment Test Center

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Due: Oct 5, 2025

Adding Sample Data

Source code:

```
20      -- ----- Equipment -----
21 • INSERT INTO Equipment (Name, Model_Number, Manufacturer, Status) VALUES
22 ('Centrifuge', 'CF-200', 'LabTech', 'Available'),
23 ('Spectrometer', 'SP-900', 'SpecCorp', 'Under Maintenance'),
24 ('Microscope', 'MS-1000', 'Optix', 'In Use');
25
26      -- ----- Client -----
27 • INSERT INTO Client (First_Name, Last_Name, Contact_Email, Phone_Number, Organization) VALUES
28 ('Aisha', 'Khan', 'aisha.khan@biotech.ca', '416-555-1010', 'BioNova Labs'),
29 ('Michael', 'Osei', 'm.osei@chemcore.com', '416-555-1020', 'ChemCore Analytics'),
30 ('Sophia', 'Martinez', 's.martinez@greeneearth.org', '416-555-1030', 'GreenEarth Research');
31
32      -- ----- Technician -----
33 • INSERT INTO Technician (First_Name, Last_Name, Certification, Contact_Email, Phone_Number) VALUES
34 ('Liam', 'Chen', 'ISO-15189', 'liam.chen@labtech.ca', '647-555-1001'),
35 ('Fatima', 'Rahman', 'ASTM-Cert', 'fatima.r@qualitylabs.com', '647-555-1002'),
36 ('Ethan', 'Brown', 'ISO-9001', 'ethan.brown@microlabs.ca', '647-555-1003');
37
38
39      -- ----- Test -----
40 • INSERT INTO Test (Test_Name, Description) VALUES
41 ('Water Purity', 'Chemical and biological analysis of water samples'),
42 ('Microbial Culture', 'Culture and count of microorganisms'),
43 ('Material Strength', 'Tensile and compression strength test');
44
45      -- ----- Lab -----
46 • INSERT INTO Lab (Lab_Name, Location, Capacity) VALUES
47 ('Main Lab', 'Building A - Floor 1', 10),
48 ('Microbiology Lab', 'Building C - Floor 3', 8),
49 ('Materials Lab', 'Building B - Floor 2', 6);
50
51      -- ----- Booking -----
52 • INSERT INTO Booking (Booking_Date, Client_ID, Lab_ID, Start_Time, End_Time, Status) VALUES
53 ('2025-10-05', 1, 1, '09:00:00', '11:00:00', 'Confirmed'),
54 ('2025-10-06', 2, 2, '13:00:00', '15:30:00', 'Pending'),
55 ('2025-10-07', 3, 3, '10:00:00', '12:00:00', 'Cancelled');
```

```

56  -- ----- Booking_Equipment -----
57  • INSERT INTO Booking_Equipment (Booking_ID, Equipment_ID) VALUES
58    (1,1),
59    (2,2),
60    (3,3);
61
62  -- ----- Test_Session -----
63  • INSERT INTO Test_Session (Test_Date, Result, Duration, Technician_ID, Test_ID) VALUES
64    ('2025-10-05', 'Pass', '01:30:00', 1, 1),
65    ('2025-10-06', 'Fail', '02:00:00', 2, 2),
66    ('2025-10-07', 'Pending', '01:45:00', 3, 3);
67
68  -- ----- TestSession_Equipment -----
69  • INSERT INTO TestSession_Equipment (Session_ID, Equipment_ID) VALUES
70    (1,1),
71    (2,2),
72    (3,3);
73
74  -- ----- Report -----
75  • INSERT INTO Report (Report_Date, Findings, Approved_By, Session_ID) VALUES
76    ('2025-10-05', 'Sample meets purity standards', 1, 1),
77    ('2025-10-06', 'Microbial count exceeded limits', 2, 2),
78    ('2025-10-07', 'Testing in progress', 3, 3);
79
80  -- ----- Invoice -----
81  • INSERT INTO Invoice (Booking_ID, Invoice_Date, Amount, Payment_Status) VALUES
82    (1, '2025-10-05', 250.00, 'Paid'),
83    (2, '2025-10-06', 180.00, 'Unpaid'),
84    (3, '2025-10-07', 200.00, 'Unpaid');
85
86  -- ----- Maintenance_Record -----
87  • INSERT INTO Maintenance_Record (Equipment_ID, Maintenance_Date, Technician_ID, Notes) VALUES
88    (1, '2025-09-10', 1, 'Rotor inspected and cleaned'),
89    (2, '2025-09-15', 2, 'Lamp replaced and calibrated'),
90    (3, '2025-09-20', 3, 'Lens alignment adjusted');
91

```

Tables:

	Equipment_ID	Name	Model_Number	Manufacturer	Status
▶	1	Centrifuge	CF-200	LabTech	Available
	2	Spectrometer	SP-900	SpecCorp	Under Maintenance
	3	Microscope	MS-1000	Optix	In Use
•	NULL	NULL	NULL	NULL	NULL

	Session_ID	Test_Date	Result	Duration	Technician_ID	Test_ID
▶	1	2025-10-05	Pass	01:30:00	1	1
	2	2025-10-06	Fail	02:00:00	2	2
	3	2025-10-07	Pending	01:45:00	3	3
★	NULL	NULL	NULL	NULL	NULL	NULL

	Report_ID	Report_Date	Findings	Approved_By	Session_ID
▶	1	2025-10-05	Sample meets purity standards	1	1
	2	2025-10-06	Microbial count exceeded limits	2	2
	3	2025-10-07	Testing in progress	3	3
★	NULL	NULL	NULL	NULL	NULL

	Invoice_ID	Booking_ID	Invoice_Date	Amount	Payment_Status
▶	1	1	2025-10-05	250.00	Paid
	2	2	2025-10-06	180.00	Unpaid
	3	3	2025-10-07	200.00	Unpaid
★	NULL	NULL	NULL	NULL	NULL

	Booking_ID	Equipment_ID
▶	1	1
	2	2
	3	3
★	NULL	NULL

	Session_ID	Equipment_ID
▶	1	1
	2	2
	3	3
★	NULL	NULL

	Maintenance_ID	Equipment_ID	Maintenance_Date	Technician_ID	Notes
▶	1	1	2025-09-10	1	Rotor inspected and cleaned
	2	2	2025-09-15	2	Lamp replaced and calibrated
	3	3	2025-09-20	3	Lens alignment adjusted
★	NULL	NULL	NULL	NULL	NULL

Queries

Query 1: List all bookings with client names

```
3 -- 1. List all bookings with client names
4 • SELECT b.Booking_ID, c.First_Name, c.Last_Name, b.Booking_Date, b.Status
5 FROM Booking b
6 JOIN Client c ON b.Client_ID = c.Client_ID;
```

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content:
	Booking_ID	First_Name	Last_Name	Booking_Date	Status
▶	1	Aisha	Khan	2025-10-05	Confirmed
	2	Michael	Osei	2025-10-06	Pending
	3	Sophia	Martinez	2025-10-07	Cancelled

Query 2: List all equipment currently “In Use”

```
3 • SELECT Equipment_ID, Name, Status
4 FROM Equipment
5 WHERE Status = 'In Use';
6
7
```

Result Grid			
		Filter Rows:	Edit:
	Equipment_ID	Name	Status
▶	3	Microscope	In Use
*	NULL	NULL	NULL

Query 3: List clients who have at least one confirmed booking

```
3 • SELECT First_Name, Last_Name
4 FROM Client
5 WHERE Client_ID IN (
6     SELECT Client_ID
7     FROM Booking
8     WHERE Status = 'Confirmed'
9 );
```

Result Grid

	First_Name	Last_Name
▶	Aisha	Khan

Query 4: Subquery - List tests that have never failed in any session

```
3 • SELECT Test_Name
4 FROM Test
5 WHERE Test_ID NOT IN (
6     SELECT Test_ID
7     FROM Test_Session
8     WHERE Result = 'Fail'
9 );
```

Result Grid

	Test_Name
▶	Water Purity
	Material Strength

Query 5 : Correlated subquery – Find the latest booking date for each client

```
3 • SELECT c.First_Name, c.Last_Name,
4      (SELECT MAX(b.Booking_Date)
5       FROM Booking b
6       WHERE b.Client_ID = c.Client_ID) AS Latest_Booking
7 FROM Client c;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	First_Name	Last_Name	Latest_Booking
▶	Aisha	Khan	2025-10-05
	Michael	Osei	2025-10-06
	Sophia	Martinez	2025-10-07

Query 6 : Window function – Rank clients by number of bookings

```
3 • SELECT c.First_Name, c.Last_Name, COUNT(b.Booking_ID) AS Total_Bookings,
4      RANK() OVER (ORDER BY COUNT(b.Booking_ID) DESC) AS Booking_Rank
5 FROM Client c
6 LEFT JOIN Booking b ON c.Client_ID = b.Client_ID
7 GROUP BY c.Client_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	First_Name	Last_Name	Total_Bookings	Booking_Rank
▶	Aisha	Khan	1	1
	Michael	Osei	1	1
	Sophia	Martinez	1	1

Result Grid

Query 7: Window function – Assign row numbers to test sessions per technician

```
2
3 • SELECT Technician_ID, Session_ID, Test_Date,
4         ROW_NUMBER() OVER (PARTITION BY Technician_ID ORDER BY Test_Date) AS Session_Number
5 FROM Test_Session;
6
7
8
9
10
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Technician_ID	Session_ID	Test_Date	Session_Number
▶	1	1	2025-10-05	1
	2	2	2025-10-06	1
	3	3	2025-10-07	1

Query 8: Window function – Divide bookings into two groups

```
2
3 • SELECT Booking_ID, Booking_Date, Client_ID,
4         NTILE(2) OVER (ORDER BY Booking_Date) AS Booking_Group
5 FROM Booking;
6
7
8
9
10
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Booking_ID	Booking_Date	Client_ID	Booking_Group
▶	1	2025-10-05	1	1
	2	2025-10-06	2	1
	3	2025-10-07	3	2

Query 9: List reports with technician names

```
2
3 • SELECT r.Report_ID, r.Findings, t.First_Name, t.Last_Name AS Technician
4 FROM Report r
5 JOIN Technician t ON r.Approved_By = t.Technician_ID;
6
7
8
9
10
11
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Report_ID	Findings	First_Name	Technician
▶	1	Sample meets purity standards	Liam	Chen
	2	Microbial count exceeded limits	Fatima	Rahman
	3	Testing in progress	Ethan	Brown

Query 10: List invoices above average amount

```
2
3 • SELECT i.Invoice_ID, i.Booking_ID, i.Amount
4 FROM Invoice i
5 JOIN (
6     SELECT AVG(Amount) AS Avg_Amount
7     FROM Invoice
8 ) avg_table ON i.Amount > avg_table.Avg_Amount;
9
10
11
12
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Invoice_ID	Booking_ID	Amount
▶	1	1	250.00

Views

- View 1:

```
1 • USE EquipmentTestCenterDB;
2
3 • CREATE OR REPLACE VIEW Equipment_Status_Summary AS
4 SELECT
5     Name AS Equipment_Name,
6     Status,
7     COUNT(*) AS Count_By_Status
8 FROM Equipment
9 GROUP BY Name, Status
10 ORDER BY Name, Status;
11
12
13 • SELECT * FROM Equipment_Status_Summary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Equipment_Name	Status	Count_By_Status
▶	Centrifuge	Available	1
	Microscope	In Use	1
	Spectrometer	Under Maintenance	1

- View 2:

```
1 • USE EquipmentTestCenterDB;
2
3 • CREATE OR REPLACE VIEW Equipment_Status_Summary AS
4 SELECT
5     Name AS Equipment_Name,
6     Status,
7     COUNT(*) AS Count_By_Status
8 FROM Equipment
9 GROUP BY Name, Status
10 ORDER BY Name, Status;
11
12
13 • SELECT * FROM Equipment_Status_Summary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Equipment_Name	Status	Count_By_Status
▶	Centrifuge	Available	1
	Microscope	In Use	1
	Spectrometer	Under Maintenance	1

- View 3:

```

3 • CREATE OR REPLACE VIEW Test_Session_Details AS
4 SELECT DISTINCT
5     ts.Session_ID,
6     t.Test_Name,
7     CONCAT(tech.First_Name, ' ', tech.Last_Name) AS Technician_Name,
8     ts.Test_Date,
9     TIME_TO_SEC(ts.Duration)/60 AS Duration_Minutes,
10    ts.Result
11 FROM Test_Session ts
12 JOIN Test t ON ts.Test_ID = t.Test_ID
13 JOIN Technician tech ON ts.Technician_ID = tech.Technician_ID
14 ORDER BY ts.Test_Date;
15
16 • SELECT * FROM Test Session Details;

```

Result Grid						
Filter Rows:						
Export:  Wrap Cell Content: 						
	Session_ID	Test_Name	Technician_Name	Test_Date	Duration_Minutes	Result
1	1	Water Purity	Liam Chen	2025-10-05	90.0000	Pass
2	2	Microbial Culture	Fatima Rahman	2025-10-06	120.0000	Fail
3	3	Material Strength	Ethan Brown	2025-10-07	105.0000	Pending